

Protein supplement

A **protein supplement** is a dietary supplement or a bodybuilding supplement, and usually comes in the form of a protein bar, protein powder, and even readily available as a protein shake. Protein supplements are usually made from whey, plant, and/or meat sources.

Protein supplements are extracts or concentrates of high protein foodstuffs, used in bodybuilding and as dietary supplements to fulfill protein intake in a lean and pure source of proteins and amino acids. They have three main variants: concentrate (food is taken and concentrated into a smaller volume with some fat and carb present), isolate (proteins and amino acids are completely isolated, mostly leaving proteins and amino acids), and hydrolyze (a protein supplement is exposed to enzymes and partially digested). Some protein supplements contain digestive enzymes as an additive for digestion and absorption. The range of grams of protein per scoop varies from brand to brand, some of the more high end protein powders reaching 30g of protein per scoop.



Various protein supplements



Effects

Metabolism

A meta-study found that in the first few weeks of strength training with untrained individuals, changes in lean body mass and muscle strength during the initial weeks of strength training are not influenced by the protein supplementation, but after the first few weeks, protein supplementation "may promote muscle hypertrophy and enhance gains in muscle strength in both untrained and trained individuals"^[1] Also, whey-protein supplementation in overweight individuals may reduce the body weight, total fat mass and risk factors for cardiovascular diseases.^[2]

When combined with strength training, protein supplementation promotes greater gains in lean body mass and muscle strength as the intensity, frequency, and duration of strength training increases.^[1] It increases the muscles' strength and size, during prolonged strength training in healthy adults.^[3] A meta-study concluded that intake of protein supplements higher than around 1.6 g/kg/day do not further improve the gains in FFM (fat free mass)^[3] "at least for younger individuals",^[3] with a confidence interval from 1.03 to 2.20^[3] so "it may be prudent to recommend ~2.2 g protein/kg/d for those seeking to maximize resistance training-induced gains in FFM".^[3]

Protein and aging

As people get older, aging reduces the effect of protein intake, and experts in the field of protein and aging recommend a protein intake between 1.2 and 2.0 g/kg/day or higher for elderly adults.^{[4][5][6][7]} Higher-than-ADA protein recommendations are especially important if they have acute or chronic diseases.^[8] Increased resistance training also helps balance aging's negative effect on muscle mass.^[3]

A study group concluded: "Older people with severe kidney disease (i.e., estimated GFR less than 30 mL/min/1.73m²), but who are not on dialysis, are an exception to this rule; these individuals may need to limit protein intake."^[8]

Timing

Past studies propose that spreading protein intake through the day (with a maximum of 30g at a time) achieves maximal total stimulation of muscle protein synthesis.^{[9][10]} However, recent studies suggest that protein intake for muscle gain can be taken either throughout the day or at one time, if that is more convenient.^[11]

There is no evidence supporting specific pre/post-workout timing for protein-supplement intakes, despite the widely held belief that pre- or post-workout protein supplementation would be more effective.^[12]

Chronic kidney disease

Nutritional status can be altered in people living with chronic kidney disease (CKD). There is moderate-certainty evidence that regular consumption of oral protein-based nutritional supplements may increase serum albumin, a protein that can be lower in people with CKD, due to increased loss in urine and malnutrition. Improvements in albumin following protein supplementation may be greater in those who require hemodialysis or who are malnourished.^[13] Pre-albumin levels and mid-arm circumference measurements may also be increased following supplementation, though the certainty of evidence is low.^[13] Although these indicate possible improvements in nutritional status, it is unclear whether protein supplements affect quality of life, life expectancy, inflammation or body composition.^[13]

Heavy metals toxicity

Consumer Reports (CR) found that protein supplements regularly carry high levels of heavy metals, which are unsafe to consume. Two-thirds of tested products from CR had lead levels exceeding safe levels from a single serving,^[14] which was defined as 0.5 micrograms/day. CR found that plant-based products had much more lead than dairy/whey protein, but those still had elevated levels of lead. Some products tested by CR also had levels of cadmium and inorganic arsenic that concerned CR.^[15]

See also

- Protein
- Protein synthesis
- Single-cell protein

References

1. Pasiakos SM, McLellan TM, Lieberman HR (January 2015). "The effects of protein supplements on muscle mass, strength, and aerobic and anaerobic power in healthy adults: a systematic review". *Sports Medicine*. **45** (1): 111–131. doi:10.1007/s40279-014-0242-2 (<https://doi.org/10.1007/s40279-014-0242-2>). PMID 25169440 (<https://pubmed.ncbi.nlm.nih.gov/25169440>). S2CID 23197226 (<https://api.semanticscholar.org/CorpusID:23197226>).
2. Wirunsawanya K, Upala S, Jaruvongvanich V, Sanguankeo A (January 2018). "Whey Protein Supplementation Improves Body Composition and Cardiovascular Risk Factors in Overweight and Obese Patients: A Systematic Review and Meta-Analysis". *Journal of the American College of Nutrition*. **37** (1): 60–70. doi:10.1080/07315724.2017.1344591 (<https://doi.org/10.1080/07315724.2017.1344591>). PMID 29087242 (<https://pubmed.ncbi.nlm.nih.gov/29087242>). S2CID 1333511 (<https://api.semanticscholar.org/CorpusID:1333511>).
3. Morton RW, Murphy KT, McKellar SR, Schoenfeld BJ, Henselmans M, Helms E, et al. (March 2018). "A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5867436>). *British Journal of Sports Medicine*. **52** (6): 376–384. doi:10.1136/bjsports-2017-097608 (<https://doi.org/10.1136/bjsports-2017-097608>). PMC 5867436 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5867436>). PMID 28698222 (<https://pubmed.ncbi.nlm.nih.gov/28698222>).
4. Wolfe RR (August 2012). "The role of dietary protein in optimizing muscle mass, function and health outcomes in older individuals". *Br J Nutr*. **108** (Suppl 2): S88–93. doi:10.1017/S0007114512002590 (<https://doi.org/10.1017/S0007114512002590>). PMID 23107552 (<https://pubmed.ncbi.nlm.nih.gov/23107552>).
5. Wolfe RR, Miller SL, Miller KB (October 2008). "Optimal protein intake in the elderly". *Clin Nutr*. **27** (5): 675–684. doi:10.1016/j.clnu.2008.06.008 (<https://doi.org/10.1016/j.clnu.2008.06.008>). PMID 18819733 (<https://pubmed.ncbi.nlm.nih.gov/18819733>).
6. Volpi E, Campbell WW, Dwyer JT, Johnson MA, Jensen GL, Morley JE, Wolfe RR (June 2013). "Is the optimal level of protein intake for older adults greater than the recommended dietary allowance?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3660117>). *J Gerontol A Biol Sci Med Sci*. **68** (6): 677–681. doi:10.1093/gerona/gls229 (<https://doi.org/10.1093/gerona/gls229>). PMC 3660117 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3660117>). PMID 23183903 (<https://pubmed.ncbi.nlm.nih.gov/23183903>).
7. Baum JI, Kim IY, Wolfe RR (June 2016). "Protein Consumption and the Elderly: What Is the Optimal Level of Intake?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4924200>). *Nutrients*. **8** (6): 359. doi:10.3390/nu8060359 (<https://doi.org/10.3390/nu8060359>). PMC 4924200 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4924200>). PMID 27338461 (<https://pubmed.ncbi.nlm.nih.gov/27338461>).
8. Bauer J, Biolo G, Cederholm T, Cesari M, Cruz-Jentoft AJ, Morley JE, Phillips S, Sieber C, Stehle P, Teta D, Visvanathan R, Volpi E, Boirie Y (August 2013). "Evidence-based recommendations for optimal dietary protein intake in older people: a position paper from the PROT-AGE Study Group" (<https://doi.org/10.1016/j.jamda.2013.05.021>). *J Am Med Dir Assoc*. **14** (8): 542–59. doi:10.1016/j.jamda.2013.05.021 (<https://doi.org/10.1016/j.jamda.2013.05.021>). PMID 23867520 (<https://pubmed.ncbi.nlm.nih.gov/23867520>).
9. Paddon-Jones D, Rasmussen BB (January 2009). "Dietary protein recommendations and the prevention of sarcopenia" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2760315>). *Curr Opin Clin Nutr Metab Care*. **12** (1): 86–90. doi:10.1097/MCO.0b013e32831cef8b (<https://doi.org/10.1097/MCO.0b013e32831cef8b>). PMC 2760315 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2760315>). PMID 19057193 (<https://pubmed.ncbi.nlm.nih.gov/19057193>).

10. Layman DK (March 2009). "Dietary Guidelines should reflect new understandings about adult protein needs" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2666737>). *Nutr Metab (Lond)*. **6**: 12. doi:10.1186/1743-7075-6-12 (<https://doi.org/10.1186%2F1743-7075-6-12>). PMC 2666737 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2666737>). PMID 19284668 (<https://pubmed.ncbi.nlm.nih.gov/19284668>).
11. Deutz NE, Wolfe RR (April 2013). "Is there a maximal anabolic response to protein intake with a meal?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3595342>). *Clin Nutr*. **32** (2): 309–313. doi:10.1016/j.clnu.2012.11.018 (<https://doi.org/10.1016%2Fj.clnu.2012.11.018>). PMC 3595342 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3595342>). PMID 23260197 (<https://pubmed.ncbi.nlm.nih.gov/23260197>).
12. Schoenfeld BJ, Aragon AA, Krieger JW (December 2013). "The effect of protein timing on muscle strength and hypertrophy: a meta-analysis" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3879660>). *Journal of the International Society of Sports Nutrition*. **10** (1): 53. doi:10.1186/1550-2783-10-53 (<https://doi.org/10.1186%2F1550-2783-10-53>). PMC 3879660 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3879660>). PMID 24299050 (<https://pubmed.ncbi.nlm.nih.gov/24299050>).
13. Mah, Jia Yee; Choy, Suet Wan; Roberts, Matthew A; Desai, Anne Marie; Corken, Melissa; Gwini, Stella M; McMahon, Lawrence P (2020-05-11). Cochrane Kidney and Transplant Group (ed.). "Oral protein-based supplements versus placebo or no treatment for people with chronic kidney disease requiring dialysis" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7212094>). *Cochrane Database of Systematic Reviews*. **5** (5) CD012616. doi:10.1002/14651858.CD012616.pub2 (<https://doi.org/10.1002%2F14651858.CD012616.pub2>). PMC 7212094 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7212094>). PMID 32390133 (<https://pubmed.ncbi.nlm.nih.gov/32390133>).
14. Liao, Sharon (2026). "CR's Guide to Popular Supplements". *Consumer Reports*. Vol. 91, no. 1 (January-February). p. 30. *Paper copy consulted*.
15. Martineau, Paris (October 14, 2025). "Protein Powders and Shakes Contain High Levels of Lead" (<https://www.consumerreports.org/lead/protein-powders-and-shakes-contain-high-levels-of-lead-a4206364640/>). *Consumer Reports*. Retrieved October 14, 2025.